Complete proofs
Infinite NPA depth at minimal semidefinite complexity

Anubhav Chaturvedi

28 October 2026

Let $M = \langle A_0\rangle + \langle B_0\rangle$ and $S = \langle A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1\rangle$ in the two-input, two-output Bell scenario. We prove that their full quantum region $\mathcal{W} = \{(M, S) : p \in \mathcal{Q}_{222}\}$ has an explicit affine semidefinite lift with blocks of orders five and four and has semidefinite extension degree two, although every fixed NPA level projects strictly outside it.

The finite lift rests on equal measurement overlaps: in the nonclassical range, fixing their product leaves the determinant minimized by their smallest sum. Part I develops this reduction, physical rank-one witnesses, the two-strategy realization, the actual membership matrices, all signed asymmetric branches and the thirty-two relabelled projections. Part II gives the full NPA obstruction in five connected steps, from the quantum boundary to the polynomial limit of a finite certificate.

I. The finite representation

F1 Why equal measurement overlaps suffice

The region is determined by its supporting inequalities. For $H = h\,\mathrm{CHSH} + a(A_0 + B_0)$, with $h, a \in \mathbb{R}$, write

$$\mathcal{C} = \{(\lambda, a, h) : \lambda I - H \succeq 0 \text{ for every quantum realization}\}.$$

Jordan's lemma, after projective dilation, reduces this condition to real two-qubit blocks. Put

$$A_0 = B_0 = Z, \quad A_1 = cZ + rX, \quad B_1 = dZ + sX, \qquad r^2 = 1 - c^2,\ s^2 = 1 - d^2.$$

We may take $r, s \geq 0$ by local Z conjugations. Endpoints include the one-dimensional blocks. Complex state vectors cause no further condition, because the Bell matrix is real symmetric.

Set $H_* = |h|$, $b = |a|$ and $L = 2H_* + 2b$. Deterministic assignments attain L: choose $A_0 = B_0 = \mathrm{sign}(a)$, and take $A_1 = B_1$ equal to that sign for $h \geq 0$, or its negative for $h < 0$. At $a = 0$, either sign works. Therefore every admissible λ satisfies $\lambda \geq L$.

The determinant controls positivity. With $\Omega = 1 + c + d - cd \in [-2, 2]$, the coefficients in

$$\det(uI + \lambda I - H) = u^4 + e_1u^3 + e_2u^2 + e_3u + e_4$$

are

$$e_1 = 4\lambda, \qquad e_2 = 6\lambda^2 - 4a^2 - 8h^2,$$
$$e_3 = 4\lambda^3 - 8\lambda a^2 - 16\lambda h^2 - 8ha^2\Omega, \qquad e_4 = \det(\lambda I - H).$$

The first three are nonnegative for $\lambda \geq L$. For e_3, bound its last term below by $-16H_*b^2$. The resulting cubic is increasing for $\lambda \geq L$, and at L equals $16b(b + 2H_*)^2$. Consequently

$$\lambda I - H \succeq 0 \quad \Longleftrightarrow \quad e_4 \geq 0.$$

Indeed, a negative eigenvalue would produce a positive root of a polynomial whose coefficients are all nonnegative. This argument includes zero determinants and multiple zero eigenvalues.

At fixed product, minimize the sum. Put $x = (1-c)/2$, $y = (1-d)/2$, and denote the determinant by $D(x,y)$. Direct expansion gives

$$\begin{aligned} D(x,y) &= D_0 + 64h^2(4h^2-a^2)xy(x+y) \\ &\quad + (32\lambda h a^2 + 128h^2a^2 - 256h^4)xy - 256h^4x^2y^2, \\ D_0 &= [(\lambda-2h)^2 - 4a^2](\lambda+2h)^2. \end{aligned}$$

Only $x+y$ varies when xy is fixed. Arithmetic–geometric mean gives $x+y \geq 2\sqrt{xy}$, and therefore

$$D(x,y) - D(\sqrt{xy},\sqrt{xy}) = 64h^2(4h^2-a^2)xy(\sqrt{x}-\sqrt{y})^2. \tag{F1}$$

For $|a| \leq |h|$, the right side is nonnegative. The equal-overlap point remains in the measurement square, so positivity there implies positivity at every original pair c, d, including the endpoints.

The remaining directions are classical. The range $b \geq H_*$ has exact support value L. To prove this, set $C_B = (B_0+B_1)/2$, $D_B = (B_0-B_1)/2$. Then

$$4I - (\mathrm{CHSH} + 2A_0) = (I - C_B - A_1D_B)^2 + \frac{1}{4}\sum_{y=0}^{1}[(I-A_0)(I+B_y)]^2 \succeq 0. \tag{F2}$$

Here $C_B^2 + D_B^2 = I$ and $C_BD_B + D_BC_B = 0$. Party exchange and output flips give $\sigma\mathrm{CHSH} + 2\tau A_0 \leq 4I$ and $\sigma\mathrm{CHSH} + 2\tau B_0 \leq 4I$, for either sign σ, τ. Average these inequalities, multiply by H_*, and add $(b-H_*)[2I - \tau(A_0+B_0)] \succeq 0$. The resulting bound is $H \leq LI$, attained by the deterministic strategies above. This also covers $h = 0$, the critical line, and both coefficient signs.

The quartic matrix. On $c = d$, the Bell basis

$$\phi_\pm = (|00\rangle \pm |11\rangle)/\sqrt{2}, \qquad \psi_\pm = (|01\rangle \pm |10\rangle)/\sqrt{2}$$

splits off ψ_-, with eigenvalue $-2hc$. In the basis (ϕ_+, ϕ_-, ψ_+), the remaining block is

$$H_3(c) = \begin{pmatrix} 2hc & 2a & 0 \\ 2a & 2h(1+c-c^2) & 2h(1-c)\sqrt{1-c^2} \\ 0 & 2h(1-c)\sqrt{1-c^2} & -2h(1+c-c^2) \end{pmatrix}.$$

Use $c = (1-t^2)/(1+t^2)$ and multiply its slack by $(1+t^2)^2$. Writing

$$p_0 = (1+t^2)^2, \qquad p_1 = 1-t^4, \qquad p_2 = 1+4t^2-t^4,$$

the resulting quartic is

$$K_{h,a,\lambda}(t) = \begin{pmatrix} \lambda p_0 - 2hp_1 & -2ap_0 & 0 \\ -2ap_0 & \lambda p_0 - 2hp_2 & -8ht^3 \\ 0 & -8ht^3 & \lambda p_0 + 2hp_2 \end{pmatrix}. \tag{F3}$$

Finite $t \geq 0$ covers $-1 < c \leq 1$; $c = -1$ follows by continuity. Negative t adds no condition because $K(-t) = \mathrm{diag}(1,1,-1)K(t)\,\mathrm{diag}(1,1,-1)$.

Conversely, $K(t) \succeq 0$ for all real t forces the local bound. Its upper-left block at $t = 0$ gives $\lambda \geq 2h + 2|a|$; the leading coefficient at $t = \infty$ gives $\lambda \geq -2h + 2|a|$. Thus $\lambda \geq L$, and the omitted scalar sector is controlled. Equation (F1) handles the low-tilt range; Eq. (F2) handles the rest. We have proved, for all real coefficients,

$$(\lambda, a, h) \in \mathcal{C} \quad \Longleftrightarrow \quad K_{h,a,\lambda}(t) \succeq 0 \quad \text{for every } t \in \mathbb{R}. \tag{F4}$$

A fixed affine Gram lift. Every PSD real univariate matrix polynomial is a polynomial matrix sum of squares, including identically singular ones [24]. Since the diagonal entries of K have degree at most four, every square factor has degree at most two: highest-degree real squares cannot cancel. Average a Gram representation under the involution above. Its two parity blocks use precisely

$$Z_+ = (\xi_1, \xi_2, t\xi_3, t^2\xi_1, t^2\xi_2)^T, \qquad Z_- = (\xi_3, t\xi_1, t\xi_2, t^2\xi_3)^T.$$

Thus $\xi^T K(t)\xi = Z_+^T G Z_+ + Z_-^T J Z_-$, with $G \succeq 0$ of order five and $J \succeq 0$ of order four. The sixteen coefficient equations are

$$\begin{aligned}
g_{11} &= \lambda - 2h, & g_{12} &= -2a,\\
g_{22} &= \lambda - 2h, & j_{11} &= \lambda + 2h,\\
g_{13} + j_{12} &= 0, & g_{23} + j_{13} &= 0,\\
2g_{14} + j_{22} &= 2\lambda, & g_{15} + g_{24} + j_{23} &= -4a,\\
2g_{25} + j_{33} &= 2\lambda - 8h, & g_{33} + 2j_{14} &= 2\lambda + 8h,\\
g_{34} + j_{24} &= 0, & g_{35} + j_{34} &= -8h,\\
g_{44} &= \lambda + 2h, & g_{45} &= -2a,\\
g_{55} &= \lambda + 2h, & j_{44} &= \lambda - 2h.
\end{aligned} \tag{F5}$$

These equations include zero coefficients, lower polynomial degrees and every rank-deficient case. Setting $h = 1$ gives the common-tilt epigraph for all real a; its restriction to $0 \le a \le 1$ additionally requires those two scalar inequalities.

F2 Explicit membership matrices for the quantum region

Introduce sixteen real variables $z_1, \ldots, z_{16}$ and define

$$\begin{aligned}
U(z) &= \begin{pmatrix}
z_1 & z_2/2 & z_5/2 & z_7 & z_8/2\\
z_2/2 & z_3 & z_6/2 & z_8/2 & z_9\\
z_5/2 & z_6/2 & z_{10} & z_{11}/2 & z_{12}/2\\
z_7 & z_8/2 & z_{11}/2 & z_{13} & z_{14}/2\\
z_8/2 & z_9 & z_{12}/2 & z_{14}/2 & z_{15}
\end{pmatrix},\\
V(z) &= \begin{pmatrix}
z_4 & z_5/2 & z_6/2 & z_{10}\\
z_5/2 & z_7 & z_8/2 & z_{11}/2\\
z_6/2 & z_8/2 & z_9 & z_{12}/2\\
z_{10} & z_{11}/2 & z_{12}/2 & z_{16}
\end{pmatrix}.
\end{aligned} \tag{F6}$$

The exact membership test is

$$(M, S) \in \mathcal{W} \quad \Longleftrightarrow \quad \text{there exists } z \in \mathbb{R}^{16} \text{ with } U(z) \succeq 0, \quad V(z) \succeq 0,$$

and

$$\begin{aligned}
1 &= z_1 + z_3 + z_4 + 2z_7 + 2z_9 + 2z_{10} + z_{13} + z_{15} + z_{16},\\
M &= 2z_2 + 4z_8 + 2z_{14},\\
S &= 2z_1 + 2z_3 - 2z_4 + 8z_9 - 8z_{10} + 8z_{12} - 2z_{13} - 2z_{15} + 2z_{16}.
\end{aligned} \tag{F7}$$

We first build the membership variables from actual states. Use $u, v \ge 0$, $u^2 + v^2 = 1$, and a real unit vector $\xi = (\xi_1, \xi_2, \xi_3)$ in the triplet basis (ϕ_+, ϕ_-, ψ_+). The measurement overlap is $c = u^2 - v^2$, with sine $2uv$. Define

$$r_+ = (u^2\xi_1, u^2\xi_2, uv\xi_3, v^2\xi_1, v^2\xi_2)^T, \qquad r_- = (u^2\xi_3, uv\xi_1, uv\xi_2, v^2\xi_3)^T.$$

The rank-one matrices $U = r_+ r_+^T$, $V = r_- r_-^T$ have precisely the shared entries in Eq. (F6). Explicitly, choose

$$\begin{aligned} z = (&u^4\xi_1^2,\ 2u^4\xi_1\xi_2,\ u^4\xi_2^2,\ u^4\xi_3^2,\\ &2u^3v\xi_1\xi_3,\ 2u^3v\xi_2\xi_3,\ u^2v^2\xi_1^2,\ 2u^2v^2\xi_1\xi_2,\\ &u^2v^2\xi_2^2,\ u^2v^2\xi_3^2,\ 2uv^3\xi_1\xi_3,\ 2uv^3\xi_2\xi_3,\\ &v^4\xi_1^2,\ 2v^4\xi_1\xi_2,\ v^4\xi_2^2,\ v^4\xi_3^2). \end{aligned} \tag{F8}$$

Substitution into Eq. (F7) gives $\ell(z) = 1$ and

$$\begin{aligned} M &= 4\xi_1\xi_2, \\ S &= 2(u^4 - v^4)\xi_1^2 + 2(u^4 + 4u^2v^2 - v^4)(\xi_2^2 - \xi_3^2) + 16uv^3\xi_2\xi_3. \end{aligned}$$

These are the state's physical expectations. The homogeneous variables include both endpoints without division or limits.

Let $\mathcal{T}$ be this set of real triplet images. It is compact and path-connected, being a continuous image of a quarter-circle times S^2. The singlet contributes only $(0, -2c)$, and

$$(0, -2c) = \frac{1+c}{2}(0, -2) + \frac{1-c}{2}(0, 2).$$

Both endpoints belong to $\mathcal{T}$: choose $\xi = (0, 0, 1)$ and respectively $(u, v) = (1, 0), (0, 1)$. Since equal overlaps attain every support value, $\operatorname{conv}\mathcal{T}$ and $\mathcal{W}$ have the same supporting inequalities. They are compact convex sets, hence

$$\mathcal{W} = \operatorname{conv}\mathcal{T}. \tag{F9}$$

Convex mixtures of the displayed witnesses therefore admit every point of $\mathcal{W}$ into the SDP, including the boundary.

For the other inclusion, write Eq. (F5) as $\mathcal{A}(G, J) = B(\lambda, a, h)$ in the displayed row order. Then $(U(z), V(z)) = \mathcal{A}^* z$ and $B^T z = (\ell(z), -M(z), -S(z))$. Any feasible z satisfies

$$\lambda - aM(z) - hS(z) = \operatorname{tr}[U(z)G] + \operatorname{tr}[V(z)J] \geq 0$$

for every valid supporting inequality, by Eq. (F4) and its Gram representation. Thus its coordinates belong to $\mathcal{W}$. This proves the membership test without invoking dual attainment.

Two strategies suffice. Every point of the convex hull of a compact connected set in the plane lies on a segment joining two points of the set. Here is a direct proof. Take $p \in \operatorname{conv}\mathcal{T} \setminus \mathcal{T}$ and project $\mathcal{T}$ radially onto the unit circle about p. Its image is compact and connected. If it contains no antipodal pair, it is an arc of length less than π, contained in an open semicircle. The midpoint direction then strictly separates p from $\mathcal{T}$, a contradiction. Thus $x - p = rn$ and $y - p = -sn$ for some $x, y \in \mathcal{T}$ and $r, s > 0$, and $p = (sx + ry)/(r + s)$.

Applied to Eq. (F9), this shows that every $(M, S) \in \mathcal{W}$ is realized by a mixture of at most two real pure two-qubit strategies, each with equal local overlaps. Their witnesses give simultaneously $\operatorname{rank} U, \operatorname{rank} V \leq 2$. If the point is extreme, it belongs to $\mathcal{T}$ and both ranks can be at most one. These rank bounds describe available witnesses; no rank constraint is added to the convex SDP.

Compactness and the dual formulation. The normalized feasible auxiliary set is also compact. The following argument gives an alternative dual proof of membership and can be useful when optimizing over the lift. For the reverse implication, the dual image

$$\mathcal{D} = \{B^T z : U(z) \succeq 0,\ V(z) \succeq 0\}$$

is closed. Indeed, the positive definite matrices

$$G_0 = \operatorname{diag}(1, 1, 2, 1, 1), \qquad J_0 = \operatorname{diag}(1, 2, 2, 1)$$

satisfy $\mathcal{A}(G_0, J_0) = B(1, 0, 0)$, and hence $\ell(z) = \mathrm{tr}[U(z)G_0] + \mathrm{tr}[V(z)J_0]$. If $B^T z$ converges, this weighted trace bounds both PSD matrices. Every z_i occurs in their entries, so z is bounded and has a convergent subsequence. The limit is feasible. The strictly feasible pair (G_0, J_0) licenses the conic intersection-dual rule, giving $\mathcal{C}^* = \overline{\mathcal{D}} = \mathcal{D}$. Finally, the compact convex set $\mathcal{W}$ is the intersection of its support half-spaces, so $(M, S) \in \mathcal{W}$ exactly when $(1, -M, -S) \in \mathcal{C}^*$. This proves the membership test. The normalized feasible z-set is itself compact by the same trace argument.

F3 Minimal block order and the NPA comparison

The set $\mathcal{W}$ is compact and convex. The explicit lift makes it semialgebraic. Scheiderer's planar theorem therefore gives $\mathrm{sxdeg}(\mathcal{W}) \leq 2$ [25]. A short argument shows that $\mathcal{W}$ is not a polygon.

Let $f(a) = \max_{\mathcal{W}}(S + aM)$, and $w = 2\sqrt{2} = f(0)$. The identity

$$\mathrm{CHSH}(c, d)^2 = 4I + 4rs\, Y \otimes Y$$

shows that attaining w requires $r = s = 1$. At $c = d = 0$, the top state is $v = \cos(\pi/8)\phi_- + \sin(\pi/8)\psi_+$, with both reduced states $I/2$. Thus every CHSH-maximizing behavior has $M = 0$. For maximizers p_a of $f(a)$, compactness then gives $M(p_a) \to 0$ as $a \downarrow 0$, and

$$0 \leq \frac{f(a) - w}{a} \leq M(p_a) \longrightarrow 0.$$

Yet $f(a) > w$ for every $a > 0$: keeping $c = d = 0$, the normalized state $(v + \varepsilon\phi_+)/\sqrt{1 + \varepsilon^2}$ gives

$$\frac{w + 4a\cos(\pi/8)\varepsilon}{1 + \varepsilon^2} > w \quad \text{for} \quad \varepsilon = \frac{2a\cos(\pi/8)}{w}.$$

A polygon would make f piecewise affine with finitely many pieces. Its zero right derivative at zero would force an initial constant piece, contradicting this strict increase. Therefore $\mathcal{W}$ is nonpolyhedral, and

$$\boxed{\mathrm{sxdeg}(\mathcal{W}) = 2.}$$

The same proof gives extension degree two for $\{(a, \lambda) : 0 \leq a \leq 1,\ \lambda \geq f(a)\}$.

The prior NPA result applies to these very support directions $h = 1$ [26]:

$$f(1 - T) = 4 - 2T + \frac{4}{3}T^3 + O(T^4), \qquad \frac{\omega_{\mathrm{NPA}}^{(L)}(1 - T) - f(1 - T)}{T^3} \longrightarrow +\infty$$

for every fixed finite level L. Consequently no fixed NPA projection onto (M, S) equals $\mathcal{W}$, while Eqs. (F6)–(F7) describe that same region exactly. The nontermination theorem is proved in Part II. The finite lift and the nonpolyhedrality argument above are independent of that proof.

F4 All signed asymmetric tilt branches

The weighted-ray argument also extends to arbitrary real tilts in

$$H_{\alpha,\beta} = \mathrm{CHSH} + \alpha A_0 + \beta B_0.$$

Put $q = |\alpha| + |\beta|$. Its local bound is $2 + q$, attained for any tilt signs. If $q \geq 2$, signed one-sided versions of Eq. (F2), convex combination on $q = 2$, and sign-adjusted marginal slacks above it prove that this is also the quantum bound.

Suppose $q < 2$ and $\lambda \geq 2 + q$. For the full Jordan slack,

$$e_2 = 6\lambda^2 - 2\alpha^2 - 2\beta^2 - 8, \qquad e_3 = 4\lambda^3 - 4\lambda(\alpha^2 + \beta^2 + 4) - 8\alpha\beta\Omega.$$

Both are nonnegative. For e_3, a lower bound is $g(\lambda) = 4\lambda^3 - 4\lambda(q^2+4) - 4q^2$. It increases on this range, and $g(2+q) = 32q + 12q^2 \geq 0$. Thus the determinant again decides PSD.

With $x = (1-c)/2$, $y = (1-d)/2$, direct expansion gives

$$\begin{aligned} D(x,y) &= D_0 + xy[C + 64Ax + 64By - 256xy], \\ A &= 4 - \alpha^2, \qquad B = 4 - \beta^2, \\ C &= 32\lambda\alpha\beta + 64\alpha^2 + 64\beta^2 - 256, \\ D_0 &= [(\lambda-2)^2 - (\alpha+\beta)^2][(\lambda+2)^2 - (\alpha-\beta)^2]. \end{aligned}$$

Both A, B are positive. Exchange parties if necessary so that $|\alpha| \geq |\beta|$, and set $\rho = A/B \in (0,1]$. For a fixed product $p = xy$, minimize $Ax + Bp/x$ on $p \leq x \leq 1$. For $p \leq \rho$, its minimizer lies on $x = z, y = \rho z$. For $p > \rho$, it lies on $x = 1$. On that edge,

$$D(1,y) = -F_-(y)F_+(y),$$

where

$$F_- = -(\lambda+\alpha)^2 + \beta^2 + 4 + 4\beta(2y-1), \qquad F_+ = (\lambda-\alpha)^2 - \beta^2 - 4 + 4\beta(2y-1).$$

Since $\lambda \pm \alpha \geq 2 + |\beta|$, we have $F_- \leq 0 \leq F_+$. Hence the complete measurement square reduces to the weighted ray, for every choice of tilt signs.

Along the ray, put $R = 4z(1-z)$, $T = 4\rho z(1-\rho z)$, and $\Omega = 2 - 4\rho z^2$. Congruence by $\mathrm{diag}(\sqrt{RT}, \sqrt{R}, \sqrt{T}, 1)$ gives

$$P(z) = \begin{pmatrix} RT(\lambda - \Omega - \alpha - \beta) & -2zRT & -2\rho zRT & RT \\ -2zRT & R(\lambda + \Omega - \alpha + \beta) & RT & 2\rho zR \\ -2\rho zRT & RT & T(\lambda + \Omega + \alpha - \beta) & 2zT \\ RT & 2\rho zR & 2zT & \lambda - \Omega + \alpha + \beta \end{pmatrix}. \quad \text{(F10)}$$

This matrix has degree at most six. The congruence is invertible on $0 < z < 1$; the original slack at both endpoints follows by continuity from that interval. The matrix interval sum-of-squares theorem gives

$$P(z) = Y_3(z)^T U Y_3(z) + z(1-z)Y_2(z)^T V Y_2(z), \qquad U \succeq 0, \quad V \succeq 0, \quad \text{(F11)}$$

with $Y_j = (I_4, zI_4, \ldots, z^j I_4)^T$, and block orders $16, 12$. One can obtain this interval theorem directly from the real-line matrix theorem: apply it to $(1+t^2)^6 P(t^2/(1+t^2))$, split its factors into even and odd parts, and divide back. Even factors become polynomials of degree at most three in z; odd factors give $\sqrt{z(1-z)}$ times polynomials of degree at most two. This argument also includes singular matrices and endpoint zeros.

For fixed α, β, coefficient matching in Eq. (F11) is affine in λ, U, V. Include $\lambda \geq 2 + q$ explicitly. This yields an exact fixed-size SDP throughout the signed subcritical diamond, including either zero-tilt axis and CHSH. The critical and exterior branches are the elementary local-bound certificates already proved. The ratio ρ still depends nonlinearly on the two independent tilts: this construction computes their support values, while the common-tilt construction above gives an affine membership lift.

F5 Thirty-two exact projections and their NPA intersections

For a sign pattern $\chi \in \{\pm 1\}^{2\times 2}$ satisfying $\prod_{ij} \chi_{ij} = -1$, and $x, y \in \{0,1\}$, set

$$M_{\chi,xy} = a_x + \chi_{xy} b_y, \qquad S_\chi = \sum_{ij} \chi_{ij} E_{ij}.$$

The quantum image of $(M_{\chi,xy}, S_\chi)$ is exactly $\mathcal{W}$. Here is an explicit physical relabelling. In the canonical observables replace

$$A'_0 = A_x, \quad A'_1 = \varepsilon_1 A_{1-x}, \qquad B'_0 = \eta_0 B_y, \quad B'_1 = \eta_1 B_{1-y},$$

where

$$\eta_0 = \chi_{xy}, \qquad \eta_1 = \chi_{x,1-y}, \qquad \varepsilon_1 = \chi_{1-x,y}/\eta_0.$$

Then $a'_0 + b'_0 = M_{\chi,xy}$. The first three CHSH coefficients are the corresponding entries of χ, and the fourth is $-\varepsilon_1\eta_1 = \chi_{1-x,1-y}$ because the product of the four entries is -1. The relabelling is invertible and preserves quantum realizability, which proves equality of the images.

These are thirty-two distinct two-dimensional subspaces of the space of Bell functionals. The marginal part identifies x, y. Equality of the correlation directions would require $\chi' = \chi$ or $\chi' = -\chi$. The latter changes the relative sign of the two nonzero marginal coefficients, so the marginal directions are not proportional. Thus only the identical pair gives the same plane.

Let $\mathcal{Q}_L$ be a fixed standard NPA relaxation and define

$$\mathcal{R}_L = \mathcal{Q}_L \cap \bigcap_{\chi,x,y} \{p : \pi_{\chi,xy}(p) \in \mathcal{W}\}, \qquad \pi_{\chi,xy}(p) = (M_{\chi,xy}, S_\chi).$$

Every quantum behavior survives, while each imposed image lies in $\mathcal{W}$:

$$\mathcal{Q}_{222} \subseteq \mathcal{R}_L \subseteq \pi^{-1}_{\chi,xy}(\mathcal{W}).$$

For any real h, α, these two inclusions imply

$$\sup_{p\in\mathcal{R}_L} [hS_\chi(p) + \alpha M_{\chi,xy}(p)] = \sup_{p\in\mathcal{Q}_{222}} [hS_\chi(p) + \alpha M_{\chi,xy}(p)].$$

Part II gives directions on the canonical plane for which the level-L support is strictly larger. Hence $\mathcal{R}_L \subsetneq \mathcal{Q}_L$ for every finite L.

The rational data file `ALL32_EXACT_PROJECTION_CONSTRAINTS.json` implements each copy of Eqs. (F6)–(F7) with the two appropriate linear forms. The added constraints use 64 PSD blocks, alternating orders five and four, total order 288, 512 auxiliary variables and 96 affine equations. The generator enumerates physical relabellings and checks the distinct maps exactly. The inclusions above establish exactness on the stated planes; they impose no claim of joint quantum realizability outside those projections.

II. The complete finite-NPA obstruction

S1 Bell values, quantum models, and finite moment certificates

Alice and Bob choose $x, y \in \{0, 1\}$ and obtain $a, b \in \{\pm 1\}$. A valid correlation is a nonnegative, normalized, no-signaling table $p(a, b|x, y)$. Its four marginal expectations and four correlators determine it:

$$p(a, b|x, y) = \tfrac{1}{4}[1 + a\langle A_x\rangle + b\langle B_y\rangle + ab\langle A_x B_y\rangle]. \tag{S1}$$

Here the expectations are the averages of a, b, and ab under the corresponding marginals or joint distribution. For $0 \le \alpha < 1$, set

$$h_\alpha = A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1 + \alpha(A_0 + B_0), \qquad \beta_\alpha(p) = \langle h_\alpha\rangle_p. \tag{S2}$$

The CHSH bound 2 and the two marginal bounds give the local maximum $\omega_{\mathrm{L}}(\alpha) = 2+2\alpha$, attained when all outputs are $+1$. Write $\omega_{\mathrm{Q}}(\alpha) = \max_{p\in\mathcal{Q}} \beta_\alpha(p)$ and $\Delta_\alpha = \omega_{\mathrm{Q}}(\alpha) - \omega_{\mathrm{L}}(\alpha)$.

The quantum set $\mathcal{Q}$ consists of $p(a, b|x, y) = \mathrm{Tr}[\rho(E_{a|x} \otimes F_{b|y})]$, with a density operator ρ and local binary POVMs. A common local Naimark dilation preserves all probabilities and makes both settings projective [11]; the commuting-model version follows from Proposition 3.4(b) of Ref. [12]. Their differences are self-adjoint reflections, $A_x^2 = B_y^2 = I$, with projectors $(I + aA_x)/2$, $(I + bB_y)/2$.

Jordan's lemma decomposes each finite-dimensional local pair into invariant blocks of dimension at most two [13, 14, 15]. For block projections $P_{ij} = \Pi_i^A \otimes \Pi_j^B$, the weights $r_{ij} = \mathrm{Tr}(P_{ij}\rho)$ and states $P_{ij}\rho P_{ij}/r_{ij}$ give $p = \sum_{ij:r_{ij}>0} r_{ij}p_{ij}$. Conversely, correlated classical labels realize any finite mixture by local direct sums. If $\mathcal{K}_2$ denotes the two-qubit correlations with projective measurements, including trivial projectors, then $\mathcal{Q}_{\mathrm{fd}} = \mathrm{conv}\,\mathcal{K}_2$. Carathéodory's theorem requires at most nine components in the eight-dimensional correlation space. The state and projector parameter sets are compact, so the mixture set is the compact continuous image of $\Delta_8 \times \mathcal{K}_2^9$. Finite-dimensional approximation adds no tensor-product correlations. The local two-projection algebra is nuclear [16, Sec. 4, Example 7(2)]; Proposition 3.4 and Remark 3.6 of Ref. [12] therefore identify the commuting set with the tensor-product closure. Thus, with complex Hilbert spaces throughout,

$$\mathcal{Q}_{\mathrm{fd}} = \mathcal{Q} = \overline{\mathcal{Q}_{\mathrm{fd}}} = \mathcal{Q}_{\mathrm{c}} = \mathrm{conv}\,\mathcal{K}_2. \tag{S3}$$

In particular, NPA converges to $\mathcal{Q}$ here [17, Theorem 8].

To define its finite levels, use the complex group algebra of

$$G = \langle a_0, a_1, b_0, b_1 : a_x^2 = b_y^2 = e,\ a_x b_y = b_y a_x\rangle. \tag{S4}$$

Reduced words have Alice's alternating letters before Bob's alternating letters; the adjoint reverses a word and conjugates its coefficient. Let $\mathcal{W}_L$ contain the reduced words of length at most L, let $V_L = \mathrm{span}_{\mathbb{C}}\,\mathcal{W}_L$, and enumerate its words as $\boldsymbol{w}$. A Hermitian functional ℓ on words through length $2L$ defines

$$\Gamma_L(\ell) = (\ell(u^*v))_{u,v\in\mathcal{W}_L}, \qquad \omega_{\mathrm{NPA}}^{(L)}(\alpha) = \max_{\ell(e)=1,\ \Gamma_L(\ell)\succeq 0} \ell(h_\alpha), \tag{S5}$$

where h_α now uses the abstract generators. Physical moments $\ell(w) = \langle\psi|\pi(w)|\psi\rangle$ make Γ_L a Gram matrix; therefore $\omega_{\mathrm{NPA}}^{(L)} \ge \omega_{\mathrm{Q}}$. Level 1 is an observable moment bound and need not give nonnegative probabilities. We extract correlations only at levels $d \ge 2$ below.

Proposition S1 (Attainment and Gram trace bound)**.** *For every finite $L \ge 1$ and $0 \le \alpha \le 1$, both optima in*

$$\omega_{\mathrm{NPA}}^{(L)}(\alpha) = \min\{\lambda \in \mathbb{R} : \lambda e - h_\alpha = \boldsymbol{w}^* Q \boldsymbol{w},\ Q \succeq 0\} \tag{S6}$$

are attained. The Gram matrix may be real symmetric. Every such certificate satisfies

$$\mathrm{Tr}\, Q = \lambda, \qquad \|Q\| \le \lambda, \qquad \|Q^{1/2}\| \le \sqrt{\lambda}. \tag{S7}$$

Proof. The canonical trace $\tau(\sum_g c_g g) = c_e$ obeys $\tau(u^*v) = \delta_{uv}$, so $\Gamma_L(\tau) = I \succ 0$. For feasible ℓ, the two-by-two principal minors give $|\ell(u^*v)|^2 \le \ell(u^*u)\ell(v^*v) = 1$. Every word through length $2L$ is u^*v with $u, v \in \mathcal{W}_L$ (split the word into two pieces of length at most L and invert the first). Thus the feasible set is closed and bounded in a finite-dimensional space, so the maximum is attained. Strict feasibility and finiteness give dual attainment and equality by Slater's theorem [19]. The dual identity is precisely Eq. (S6): writing $Q = C^*C$ gives a sum $\sum_j p_j^* p_j$ with $p_j \in V_L$, and conversely every such sum gives a positive Gram matrix. The multiplication table and h_α have real coefficients, so averaging Q with $\overline{Q}$ preserves the identity and positivity. Applying τ gives $\lambda = \sum_{ij} Q_{ij}\tau(w_i^* w_j) = \operatorname{Tr} Q$, since $\tau(h_\alpha) = 0$; positivity gives the norm bounds. The moment bounds also imply $\omega_{\mathrm{NPA}}^{(L)}(\alpha) \le 4 + 2\alpha \le 6$. □

For comparison, $\mathcal{W}_1 = \{e, a_0, a_1, b_0, b_1\}$ already recovers the CHSH maximum $2\sqrt{2}$ [6, 7]. The list $\mathcal{W}_{1+AB} = \mathcal{W}_1 \cup \{a_x b_y\}_{x,y}$ gives tight certificates for one-sided tilted CHSH, $h_0 + \gamma a_0$ for $0 \le \gamma < 2$, and self-tests its maximizers [8, 9, 10]. Thus level $1 + AB$ is exact when either marginal tilt vanishes [1].

S2 The cubic quantum violation

On a nontrivial Jordan block local basis choices give $A_0 = B_0 = Z$, $A_1 = cZ + rX$, $B_1 = dZ + sX$, where $c, d \in [-1, 1]$, $r = \sqrt{1 - c^2}$, and $s = \sqrt{1 - d^2}$. The endpoints include one-dimensional blocks: the assignment $(A_0, A_1) = (\epsilon_0, \epsilon_1)$ occurs on the Z eigenspace ϵ_0 at $c = \epsilon_0\epsilon_1$, and similarly for Bob. Put $\alpha = 1 - T$, $\lambda_0(T) = 4 - 2T$, $u = 1 - c$, $v = 1 - d$, and $\Omega = 2 - uv$. The evaluated Bell operator is

$$H_T(c, d) = \begin{pmatrix} \Omega + 2(1-T) & us & vr & -rs \\ us & -\Omega & -rs & -vr \\ vr & -rs & -\Omega & -us \\ -rs & -vr & -us & \Omega - 2(1-T) \end{pmatrix}. \tag{S8}$$

Equation (S3) gives $\omega_{\mathrm{Q}}(1 - T) = \max_{c,d\in[-1,1]} \lambda_{\max} H_T(c, d)$, attained on two qubits. For later use, multiplication by a real vector $\eta = (1, b_1, b_2, b_3)^T$ yields the exact identity, also valid for signed r, s with $r^2 = 1 - c^2$, $s^2 = 1 - d^2$,

$$\begin{aligned} \mathcal{E}_T(\eta) &:= \langle \eta, (H_T - \lambda_0(T) I)\eta \rangle \\ &= -uv + 2usb_1 + 2vrb_2 - 2rsb_3 + (-6 + 2T + uv)(b_1^2 + b_2^2) \\ &\quad + (-4 + 4T - uv)b_3^2 - 2rsb_1b_2 - 2vrb_1b_3 - 2usb_2b_3. \end{aligned} \tag{S9}$$

The normalized vector's Bell excess is $\mathcal{E}_T(\eta)/\|\eta\|^2$.

Theorem 1 and Eqs. (18)–(19) of Ref. [1] identify $\omega_{\mathrm{Q}}(\alpha)$ as the largest real root in λ of

$$\begin{aligned} F(\lambda, \alpha) = \lambda^4 + (4 - \alpha^2)\lambda^3 + (\tfrac{11}{4}\alpha^4 - 12\alpha^2 - 4)\lambda^2 \\ + (2\alpha^6 - \alpha^4 - 20\alpha^2 - 32)\lambda + 5\alpha^6 - 21\alpha^4 + 16\alpha^2 - 32, \end{aligned} \tag{S10}$$

with common optimal cosine

$$c_*(\alpha) = \tfrac{1}{8}[3\alpha^2 - 4 + \sqrt{16 + 9\alpha^4 + 8\alpha^2(2\omega_{\mathrm{Q}}(\alpha) - 1)}]. \tag{S11}$$

At $\alpha = 1$, $F(\lambda, 1) = (\lambda - 4)(4\lambda^3 + 28\lambda^2 + 59\lambda + 32)/4$. The cubic is positive for $\lambda \ge 4$. The local lower bound and $\omega_{\mathrm{Q}}(\alpha) \le 4 + 2\alpha$ force every endpoint limit to be the root 4. Changing the tilt by T changes any Bell value by at most $2T$, so $\omega_{\mathrm{Q}}(1 - T) = 4 + O(T)$. Since

$$F(2 + 2\alpha, \alpha) = 4(\alpha - 1)^3(\alpha + 2)^4, \qquad \partial_\lambda F(4, 1) = 243,$$

the mean-value theorem at fixed $\alpha = 1 - T$ gives

$$\Delta_{1-T} = \frac{4T^3(3 - T)^4}{\partial_\lambda F(\widetilde{\lambda}_T, 1 - T)} = \frac{324}{243}T^3 + O(T^4) = \frac{4}{3}T^3 + O(T^4), \tag{S12}$$

where $\widetilde{\lambda}_T$ lies between the local and quantum values. The nonzero derivative also gives a unique analytic root near 4. Writing this root as $4 - 2T + \frac{4}{3}T^3 + c_4T^4 + c_5T^5 + O(T^6)$ and substituting into $F = 0$ gives $27(9c_4 + 4)T^4 + (243c_5 - 243c_4 - 120)T^5 + O(T^6) = 0$. Consequently

$$\omega_{\mathrm{Q}}(1 - T) = 4 - 2T + \tfrac{4}{3}T^3 - \tfrac{4}{9}T^4 + \tfrac{4}{81}T^5 + O(T^6), \tag{S13}$$

$$c_*(1 - T) = 1 - 2T + \tfrac{8}{9}T^2 + \tfrac{20}{81}T^3 + O(T^4). \tag{S14}$$

Analyticity gives constants $C, \delta > 0$ bounding each remainder by CT^k at its displayed order for $0 \le T < \delta$. In particular the quartic correction to the quantum value divided by T^4 tends to $-4/9$. Since $1 - \cos\theta = \theta^2/2 + O(\theta^4)$, the optimal angles obey $\theta_* \sim 2\sqrt{T}$; the cited self-testing theorem gives this scale for every maximizer in the canonical form.

This endpoint has a detector-loss interpretation [4, 1]. Let each detector fire independently with common probability η, independently also of settings and ideal outcomes, and assign $+1$ on no-detection. Then

$$\langle A_x B_y\rangle_\eta = \eta^2\langle A_x B_y\rangle + \eta(1-\eta)(\langle A_x\rangle + \langle B_y\rangle) + (1-\eta)^2,$$
$$\mathrm{CHSH}_\eta(p) = \eta^2\beta_\alpha(p) + 2(1-\eta)^2, \qquad \alpha = 2(1-\eta)/\eta.$$

The setting-1 marginals cancel in CHSH. The local bound becomes 2, and both the quantum violation and the NPA error are multiplied by η^2, preserving their ratio. Here $\alpha \uparrow 1$ corresponds to $\eta \downarrow 2/3$ [5]; since $T = (3\eta - 2)/\eta = (9/2)(\eta - 2/3) + O((\eta - 2/3)^2)$,

$$\eta^2\Delta_\alpha = 54(\eta - 2/3)^3 + O((\eta - 2/3)^4). \tag{S15}$$

S3 A two-parameter trial state and its Motzkin gap

Set $T = t^2$ and vary the two angles independently:

$$A_0 = B_0 = Z, \quad A_1 = \cos(tx)Z + \sin(tx)X, \quad B_1 = \cos(ty)Z + \sin(ty)X, \qquad x, y \in \mathbb{R}. \tag{S16}$$

Write $H(t; x, y)$ for their Bell operator. A certificate must hold for every (x, y), although optimal angles approach $(2, 2)$. In what follows $O_K(t^k)$ bounds the scalar, Euclidean, or operator norm uniformly on each compact $K \subset \mathbb{R}^2$ by $C_K|t|^k$ for $0 < |t| < \delta_K$.

At $t = 0$, $H = \mathrm{diag}(4, -2, -2, 0)$; the gaps from $|00\rangle$ are $6, 6, 4$. Its couplings to the other basis vectors are $x^2yt^3/2 + O_K(t^5)$, $xy^2t^3/2 + O_K(t^5)$, and $-xyt^2 + O_K(t^4)$. To cancel the gap before order t^6, take $\zeta_t = (1, At^3, Bt^3, Ct^2)^T$ and $\Lambda_t = 4 - 2t^2 + 4t^6/3$. Equation (S9) gives the leading gap $(4C + xy)^2t^4/4$. Hence choose $C = -xy/4$. The next coefficient is

$$\langle\zeta_t, (\Lambda_t I - H(t))\zeta_t\rangle = t^6\Big[6(A - x^2y/12)^2 + 6(B - xy^2/12)^2 + \tfrac{4}{3} + \frac{x^4y^2 + x^2y^4}{48} - \frac{x^2y^2}{4}\Big] + O_K(t^8). \tag{S17}$$

Indeed, before completing the two squares their contribution is $6A^2 - Ax^2y + 6B^2 - Bxy^2$ and the sixth-degree coefficient is $1/16$; subtracting $1/24$ leaves $1/48$. Thus the polynomial vector

$$\xi_t = (1, t^3x^2y/12, t^3xy^2/12, -t^2xy/4)^T, \qquad M(X, Y) = X^4Y^2 + X^2Y^4 - 3X^2Y^2 + 1 \tag{S18}$$

satisfies

$$\langle\xi_t, [\omega_{\mathrm{Q}}(1 - t^2)I - H(t; x, y)]\xi_t\rangle = \tfrac{4}{3}t^6M(x/2, y/2) + O_K(t^8). \tag{S19}$$

To check the remainder, sine and cosine have uniform Taylor remainders on K. Conjugation by $U = Z \otimes Z$ gives $H(-t) = UH(t)U$ and $\xi_{-t} = U\xi_t$, so the expectation is even. Replacing Λ_t by the exact quantum value changes it by $[-4t^8/9 + 4t^{10}/81 + O(t^{12})]\|\xi_t\|^2 = O_K(t^8)$, since

$$\|\xi_t\|^2 = 1 + x^2y^2t^4/16 + (x^4y^2 + x^2y^4)t^6/144. \tag{S20}$$

Equivalently, ξ_t is the top eigenvector, with first component one, through order three: its leading component equations are $6A = x^2y/2$, $6B = xy^2/2$, $4C = -xy$. The simple eigenvalue at $t = 0$ and its fixed spectral gap give an $O_K(t^4)$ vector remainder.

The normalized state $\psi_t = \xi_t/\|\xi_t\|$ has Bell value $b_t = \langle\psi_t, H(t)\psi_t\rangle$, and

$$\frac{\omega_{\mathrm{Q}}(1-t^2) - b_t(x,y)}{\Delta_{1-t^2}} \longrightarrow M(x/2, y/2) \tag{S21}$$

locally uniformly. On either axis $\xi_t = |00\rangle$ is an eigenvector at the local bound, so this ratio is exactly 1. At $(x, y) = (2, 2)$ it tends to zero. The gap in this formula is a physical strategy's gap from the quantum bound; the NPA error will add a separate nonnegative term.

Lemma S2 (Shifted Motzkin polynomial)**.** *M is nonnegative, but $M + c$ is not a sum of real polynomial squares for any $c \geq 0$.*

Proof. Arithmetic–geometric mean gives $X^4Y^2 + X^2Y^4 + 1 \geq 3X^2Y^2$. If $M + c = \sum_j q_j^2$, the highest homogeneous terms cannot cancel, so $\deg q_j \leq 3$. On either axis the sum is the constant $1 + c$; each restricted polynomial is bounded and therefore constant. Thus $q_j = a_j + b_jXY + c_jX^2Y + d_jXY^2$. The X^2Y^2 coefficient of $\sum_j q_j^2$ is $\sum_j b_j^2 \geq 0$, contradicting -3. □

The usual half-Newton-polytope property gives the same monomial restriction: if $p = \sum_j q_j^2$, then $\mathrm{Newt}(q_j) \subseteq \frac{1}{2}\,\mathrm{Newt}(p)$ [23, Theorem 1]. The homogeneous statement gives the polynomial statement by homogenization and then setting the extra variable to one. Here $\mathrm{Newt}(M + c) = \mathrm{conv}\{(0,0), (4,2), (2,4)\}$, and its half has integer points $(0,0), (1,1), (2,1), (1,2)$, exactly the four monomials above. Independent angles are essential: the restriction $M(X, X) = (X^2 - 1)^2 + 2[X(X^2 - 1)]^2$, even after adding $c \geq 0$, is a sum of squares.

S4 Polynomial limits of fixed certificates and divergence

An evaluated certificate gives squared norms of analytic functions. To obtain polynomial square factors in its limit, we first control the Taylor remainder and then bound coefficients in one fixed polynomial space. For a fixed list $\mathcal{W}_L = (w_1, \ldots, w_N)$, let

$$z_t(x,y) = (\pi_{t,x,y}(w_i)\xi_t(x,y))_{i=1}^N \in \mathbb{C}^{4N}, \tag{S22}$$

where π evaluates Eq. (S16).

Lemma S3 (Fixed-level sum-of-squares limit)**.** *Let $t_n \downarrow 0$, $Q_n \succeq 0$ be $N \times N$ matrices with $\sup_n \mathrm{Tr}\, Q_n < \infty$, and $m \geq 0$ be an integer. If*

$$t_n^{-2m}\|(Q_n^{1/2} \otimes I_4)z_{t_n}(x,y)\|^2 \longrightarrow P(x,y) \tag{S23}$$

locally uniformly for a real polynomial P, then P is a sum of squares of real polynomials.

Proof. A sine or cosine coefficient of order k in t has angular degree at most k; the entries of ξ_t have the same property. Multiplication adds these orders. Since the word list is fixed and finite, Taylor's theorem gives, on every compact K,

$$z_t = \sum_{r=0}^{m} t^r z_r + R_{m+1}, \qquad \deg z_r \leq r, \qquad \sup_K \|R_{m+1}\| \leq C_K|t|^{m+1}. \tag{S24}$$

Put $C_n = Q_n^{1/2}$, keeping all N rows even if its rank changes. The trace bound gives $\|C_n\| \leq \sqrt{\sup_n \mathrm{Tr}\, Q_n}$. Hence the polynomial vectors of degree at most m,

$$p_n = t_n^{-m}(C_n \otimes I_4)\sum_{r=0}^{m} t_n^r z_r, \qquad v_n = t_n^{-m}(C_n \otimes I_4)z_{t_n}, \qquad \sup_K \|p_n - v_n\| = O_K(t_n), \tag{S25}$$

are uniformly bounded on $K_0 = [-1,1]^2$, since $\|v_n\|^2 \to P$ uniformly there. To bound coefficients despite the negative powers of t_n, choose $m+1$ fixed distinct nodes $s_i \in [-1,1]$. Their fixed Lagrange polynomials L_i give, for each coordinate p of p_n,

$$p(x,y) = \sum_{i,j=0}^{m} p(s_i,s_j) L_i(x) L_j(y).$$

The bounded grid values therefore bound every coefficient of p_n. One subsequence makes all finitely many complex coefficients converge, giving a polynomial vector q and locally uniform convergence $p_n \to q$. On K_0, $|\|p_n\|^2 - \|v_n\|^2| \le (\|p_n\| + \|v_n\|)\|p_n - v_n\| \to 0$, so $P = \|q\|^2$ there. This is a polynomial identity on an open set and thus on $\mathbb{R}^2$. Splitting each coordinate of q into real and imaginary parts expresses P as real polynomial squares. □

Theorem S4. *For every fixed finite $L \ge 1$,*

$$\lim_{\alpha\uparrow 1} \frac{\omega_{\mathrm{NPA}}^{(L)}(\alpha) - \omega_{\mathrm{Q}}(\alpha)}{(1-\alpha)^3} = +\infty, \qquad \lim_{\alpha\uparrow 1} \frac{\omega_{\mathrm{NPA}}^{(L)}(\alpha) - \omega_{\mathrm{Q}}(\alpha)}{\Delta_\alpha} = +\infty. \tag{S26}$$

Proof. If the first divergence failed, nonnegativity of the error would give $t_n \downarrow 0$ with $\omega_{\mathrm{NPA}}^{(L)}(1-t_n^2) = \omega_{\mathrm{Q}}(1-t_n^2) + r_n t_n^6$, where, after a subsequence, $r_n \to r \ge 0$. Proposition S1 gives exact Gram certificates

$$\omega_{\mathrm{NPA}}^{(L)}(1-t_n^2)e - h_{1-t_n^2} = \boldsymbol{w}^* Q_n \boldsymbol{w}, \qquad \operatorname{Tr} Q_n = \omega_{\mathrm{NPA}}^{(L)}(1-t_n^2) \le 6.$$

Evaluate on the observables and ξ_{t_n}, then split the expectation into the quantum strategy gap and $r_n t_n^6 \|\xi_{t_n}\|^2$. Equations (S19) and (S20) give, locally uniformly,

$$t_n^{-6} \|(Q_n^{1/2} \otimes I_4) z_{t_n}\|^2 \longrightarrow \tfrac{4}{3} M(x/2, y/2) + r. \tag{S27}$$

Lemma S3 with $m = 3$ makes this polynomial a sum of real squares. Substituting $(x,y) = (2X, 2Y)$ and multiplying by $3/4$ would express $M(X,Y) + 3r/4$ as such a sum, contrary to Lemma S2. The second limit follows from $\Delta_\alpha/(1-\alpha)^3 \to 4/3$. □

The absolute NPA error remains bounded. The divergence concerns its ratio to a quantum violation that vanishes cubically. For each fixed L it also gives a full interval $1 - \delta_L < \alpha < 1$ with a strict gap; δ_L may depend on L. No assertion that one fixed $\alpha < 1$ is missed at every finite level is needed.

S5 Valid correlations, arbitrary finite word lists, and a local limit

Fix any finite list $\mathcal{S} = (S_1, \dots, S_k)$ of words in the measurement projectors. Define $\mathcal{N}_{\mathcal{S}}$ to contain the valid no-signaling correlations for which a normalized Hermitian functional ℓ satisfies

$$\Gamma_{\mathcal{S}} = (\ell(S_i^* S_j))_{ij} \succeq 0, \qquad p(a,b|x,y) = \ell(E_{a|x} F_{b|y}), \tag{S28}$$

together with the identities generated by $E_{a|x}^* = E_{a|x}$, $E_{a|x} E_{a'|x} = \delta_{aa'} E_{a|x}$, $\sum_a E_{a|x} = I$, the analogous Bob relations, and $[E_{a|x}, F_{b|y}] = 0$. The functional is defined on all finitely many words needed for these constraints and the observed probabilities; $\ell(W^*) = \overline{\ell(W)}$. If a moment is absent from the positive matrix it is free apart from the stated identities. This definition includes standard finite NPA lists; additional localizing matrices, state- or operator-optimality constraints, or problem-specific inequalities define other relaxations.

Lemma S5. *Every level-d feasible functional with $d \ge 2$ defines a valid correlation through $E_{a|x} = (e + a a_x)/2$, $F_{b|y} = (e + b b_y)/2$ and $p(a,b|x,y) = \ell(E_{a|x} F_{b|y})$. Writing $\mathcal{N}_d$ for these extracted correlations, every fixed finite list $\mathcal{S}$ admits a finite $d \ge 2$ such that $\mathcal{N}_d \subseteq \mathcal{N}_{\mathcal{S}}$.*

Proof. $P = E_{a|x}F_{b|y}$ is a self-adjoint projector in $V_2 \subseteq V_d$, so $p = \ell(P) = \ell(P^*P) \geq 0$. Summing the projectors gives $\sum_{ab} p = \ell(e) = 1$ and $\sum_b p = \ell(E_{a|x})$, independently of y; Bob's marginal is analogous. This proves validity and preserves the moments $\langle A_x \rangle_p = \ell(a_x)$, $\langle B_y \rangle_p = \ell(b_y)$, and $\langle A_x B_y \rangle_p = \ell(a_x b_y)$.

Expand each projector word using these affine formulas. A word of D projector factors expands into observable words of length at most D; reduction only shortens them. Choose $d \geq 2$ at least the largest such length and enlarge it for any other finitely many required moments. Then $\boldsymbol{S} = T\boldsymbol{w}_d$ for a real matrix T, and

$$\Gamma_{\mathcal{S}} = T\Gamma_d T^* \succeq 0. \tag{S29}$$

The group relations imply all projector identities. Any remaining unconstrained moments admit a Hermitian linear extension on the reduced-word basis, assigning conjugate values on inverse pairs. No further positive matrix involving those moments is required. Hence the extracted correlation belongs to $\mathcal{N}_{\mathcal{S}}$. □

Theorem S6. *For every fixed finite projector-word list $\mathcal{S}$ as above, $\mathcal{Q} \subsetneq \mathcal{N}_{\mathcal{S}}$. Moreover, every neighborhood of $p_{\mathrm{loc}}(a,b|x,y) = \delta_{a,1}\delta_{b,1}$ contains a correlation in $\mathcal{N}_{\mathcal{S}} \setminus \mathcal{Q}$.*

Proof. Quantum moments satisfy all the constraints. Choose $d \geq 2$ from the lemma and a primal optimizer at each α, extracting $p_{d,\alpha} \in \mathcal{N}_d \subseteq \mathcal{N}_{\mathcal{S}}$ with $\beta_\alpha(p_{d,\alpha}) = \omega^{(d)}_{\mathrm{NPA}}(\alpha)$. For α sufficiently close to one, Theorem S4 puts this value strictly above $\omega_{\mathrm{Q}}(\alpha)$, proving strict containment. Set

$$e_{d,\alpha} = \omega^{(d)}_{\mathrm{NPA}}(\alpha) - \omega_{\mathrm{Q}}(\alpha), \qquad \lambda_\alpha = \frac{2\Delta_\alpha}{\Delta_\alpha + e_{d,\alpha}}, \qquad q_\alpha = (1-\lambda_\alpha)p_{\mathrm{loc}} + \lambda_\alpha p_{d,\alpha}. \tag{S30}$$

Since $e_{d,\alpha}/\Delta_\alpha \to \infty$, the weight is in $(0,1)$ near one and tends to zero. Convexity puts q_α in $\mathcal{N}_{\mathcal{S}}$, whereas $\beta_\alpha(q_\alpha) = \omega_{\mathrm{L}}(\alpha) + 2\Delta_\alpha = \omega_{\mathrm{Q}}(\alpha) + \Delta_\alpha > \omega_{\mathrm{Q}}(\alpha)$. Thus $q_\alpha \notin \mathcal{Q}$. Every probability coordinate differs from p_{loc} by at most λ_α, proving convergence to the deterministic correlation. □

References

[1] N. Gigena, E. Panwar, G. Scala, M. Araújo, M. Farkas, and A. Chaturvedi, Self-testing tilted strategies for maximal loophole-free nonlocality, npj Quantum Inf. **11**, 82 (2025).

[2] N. Brunner, D. Cavalcanti, S. Pironio, V. Scarani, and S. Wehner, Bell nonlocality, Rev. Mod. Phys. **86**, 419–478 (2014), doi:10.1103/RevModPhys.86.419.

[3] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, Proposed experiment to test local hidden-variable theories, Phys. Rev. Lett. **23**, 880 (1969).

[4] N. Miklin, A. Chaturvedi, M. Bourennane, M. Pawłowski, and A. Cabello, Exponentially decreasing critical detection efficiency for any Bell inequality, Phys. Rev. Lett. **129**, 230403 (2022), doi:10.1103/PhysRevLett.129.230403.

[5] P. H. Eberhard, Background level and counter efficiencies required for a loophole-free Einstein–Podolsky–Rosen experiment, Phys. Rev. A **47**, R747–R750 (1993), doi:10.1103/PhysRevA.47.R747.

[6] B. S. Tsirelson, Quantum generalizations of Bell's inequality, Lett. Math. Phys. **4**, 93 (1980).

[7] M. Navascués, S. Pironio, and A. Acín, Bounding the set of quantum correlations, Phys. Rev. Lett. **98**, 010401 (2007).

[8] A. Acín, S. Massar, and S. Pironio, Randomness versus nonlocality and entanglement, Phys. Rev. Lett. **108**, 100402 (2012).

[9] C. Bamps and S. Pironio, Sum-of-squares decompositions for a family of Clauser–Horne–Shimony–Holt-like inequalities and their application to self-testing, Phys. Rev. A **91**, 052111 (2015).

[10] I. Šupić and J. Bowles, Self-testing of quantum systems: a review, Quantum **4**, 337 (2020), doi:10.22331/q-2020-09-30-337.

[11] M. Neumark, Spectral functions of a symmetric operator, Izv. Akad. Nauk SSSR Ser. Mat. **4**, 277–318 (1940).

[12] T. Fritz, Tsirelson's problem and Kirchberg's conjecture, Rev. Math. Phys. **24**, 1250012 (2012), doi:10.1142/S0129055X12500122.

[13] P. R. Halmos, Two subspaces, Trans. Am. Math. Soc. **144**, 381 (1969).

[14] Ll. Masanes, Extremal quantum correlations for N parties with two dichotomic observables per site, arXiv:quant-ph/0512100 (2005).

[15] E. Panwar, P. Pandya, and M. Wieśniak, An elegant scheme of self-testing for multipartite Bell inequalities, npj Quantum Inf. **9**, 71 (2023), doi:10.1038/s41534-023-00735-3.

[16] V. B. Scholz and R. F. Werner, Tsirelson's problem, arXiv:0812.4305 (2008).

[17] M. Navascués, S. Pironio, and A. Acín, A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations, New J. Phys. **10**, 073013 (2008).

[18] S. Pironio, M. Navascués, and A. Acín, Convergent relaxations of polynomial optimization problems with noncommuting variables, SIAM J. Optim. **20**, 2157–2180 (2010), doi:10.1137/090760155.

[19] S. Boyd and L. Vandenberghe, *Convex Optimization* (Cambridge University Press, Cambridge, 2004), Secs. 5.2.3 and 5.9.1, author's online edition.

[20] L. Vandenberghe and S. Boyd, Semidefinite programming, SIAM Rev. **38**, 49–95 (1996), doi:10.1137/1038003.

[21] T. S. Motzkin, The arithmetic-geometric inequality, in *Inequalities*, edited by O. Shisha (Academic Press, New York, 1967), pp. 205–224.

[22] D. Hilbert, Über die Darstellung definiter Formen als Summe von Formenquadraten, Math. Ann. **32**, 342 (1888).

[23] B. Reznick, Extremal PSD forms with few terms, Duke Math. J. **45**, 363 (1978).

[24] C. Hanselka and R. Sinn, Positive semidefinite univariate matrix polynomials, *Math. Z.* **292**, 83–101 (2019); Corollary 2.8.

[25] C. Scheiderer, Second-Order Cone Representation for Convex Sets in the Plane, *SIAM J. Appl. Algebra Geom.* **5**, 114–139 (2021), doi:10.1137/20M133717X; Theorem 0.1.

[26] A. Chaturvedi, *No Finite NPA Level Characterizes the Complete Quantum Set in the Simplest Bell Scenario*, project manuscript, September 2026. Its complete argument is reorganized in Part II of this companion.

[27] A. Tavakoli, A. Pozas-Kerstjens, P. Brown, and M. Araújo, Semidefinite programming relaxations for quantum correlations, Rev. Mod. Phys. **96**, 045006 (2024), doi:10.1103/RevModPhys.96.045006.